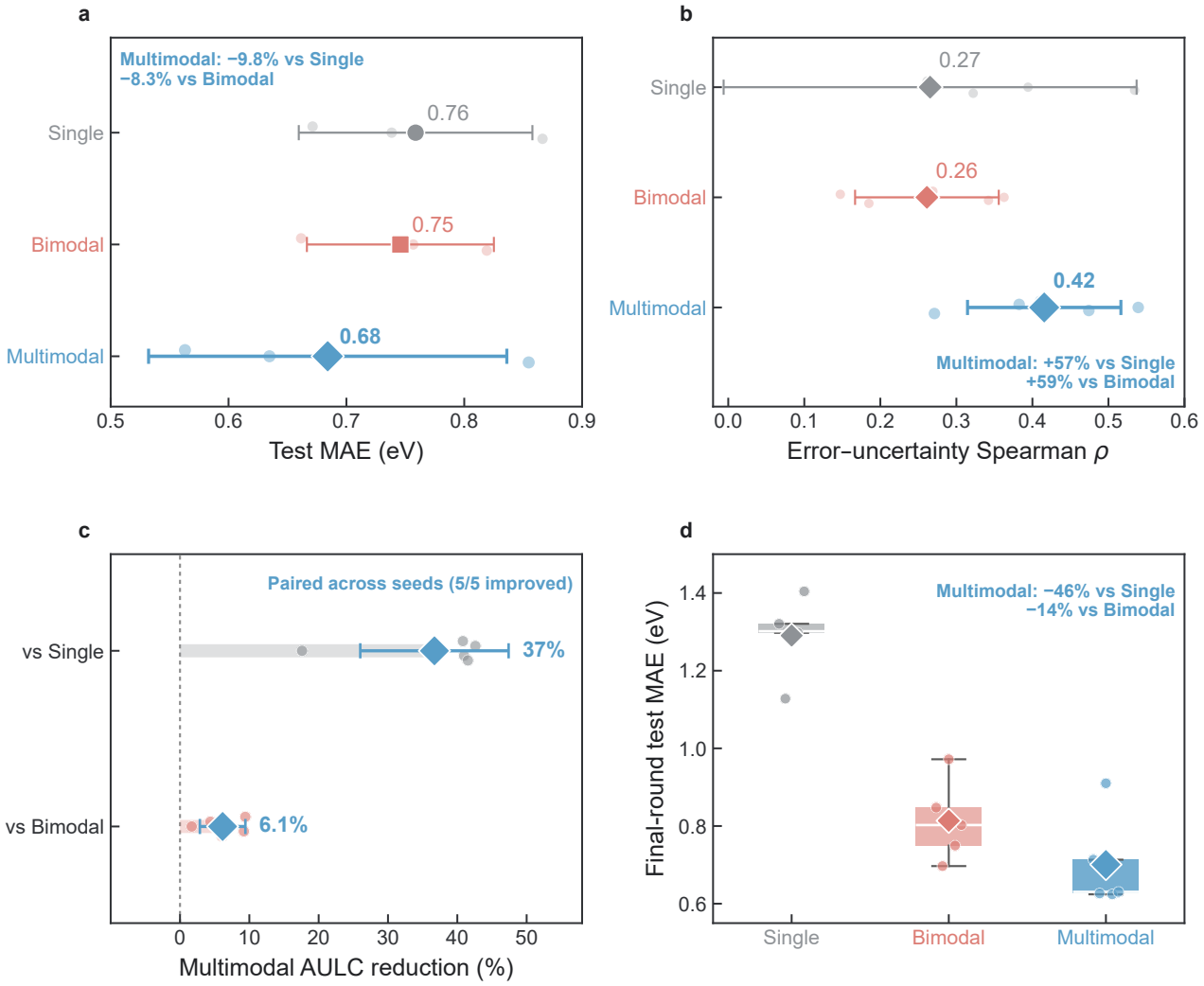

a
Multimodal: −9.8% vs Single
−8.3% vs Bimodal
Single
Bimodal
Multimodal
0.76
0.75
0.68
0.5
0.6
0.7
0.8
0.9
Test MAE (eV)
b
Single
Bimodal
Multimodal
0.27
0.26
0.42
Multimodal: +57% vs Single
+59% vs Bimodal
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Error–uncertainty Spearman ρ
c
Paired across seeds (5/5 improved)
vs Single
vs Bimodal
37%
6.1%
0
10
20
30
40
50
Multimodal AULC reduction (%)
d
Multimodal: −46% vs Single
−14% vs Bimodal
Final-round test MAE (eV)
1.4
1.2
1.0
0.8
0.6
Single
Bimodal
Multimodal